

Strong Artificial Intelligence

K Motion

Causer of Time and Space

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Technical University of Applied Sciences, Würzburg, 2026-06-26

Repetition „Language“

- Which types of languages do you know?
- What purposes of a language did we discuss?
- In which system is language a tool?
- What is semantics?
- Why is semantics a process?
- What is pragmatics?
- Which sentence has no semantics but pragmatics?

Workshop „Motion“

- What is motion?
- Which characteristics or properties would you assign to motion or movement?
- Which of your assigned properties are the most relevant?
- What basic types or categories of movement or motion would you distinguish?
- Is there a mental motion beside the physical motion? Explain.
- Which machines are build for motion? Try to find basic types.

K Motion

1. Taxonomie

How is it

2. Linear Motion

Location Change

3. Rotational Motion

Orientation Change

4. Part-Whole Logic

Disjointment

5. Rotation Logic

Triangle Paradigm

6. Position

Located Entity

7. Conclusion

Challenge

- Distance: Which distance has to be overcome?
- Velocity: How fast is the movement?
- Precision: How precise has the motion path and the target location to be defined?
- Complexity: How irregular are path and rotation of the moving part?
- Interaction: How much interaction is necessary? (Predictability)
- Intelligence: How many intelligence do you need for the motion?

1. Taxonomy



industrial robot



mobile robot

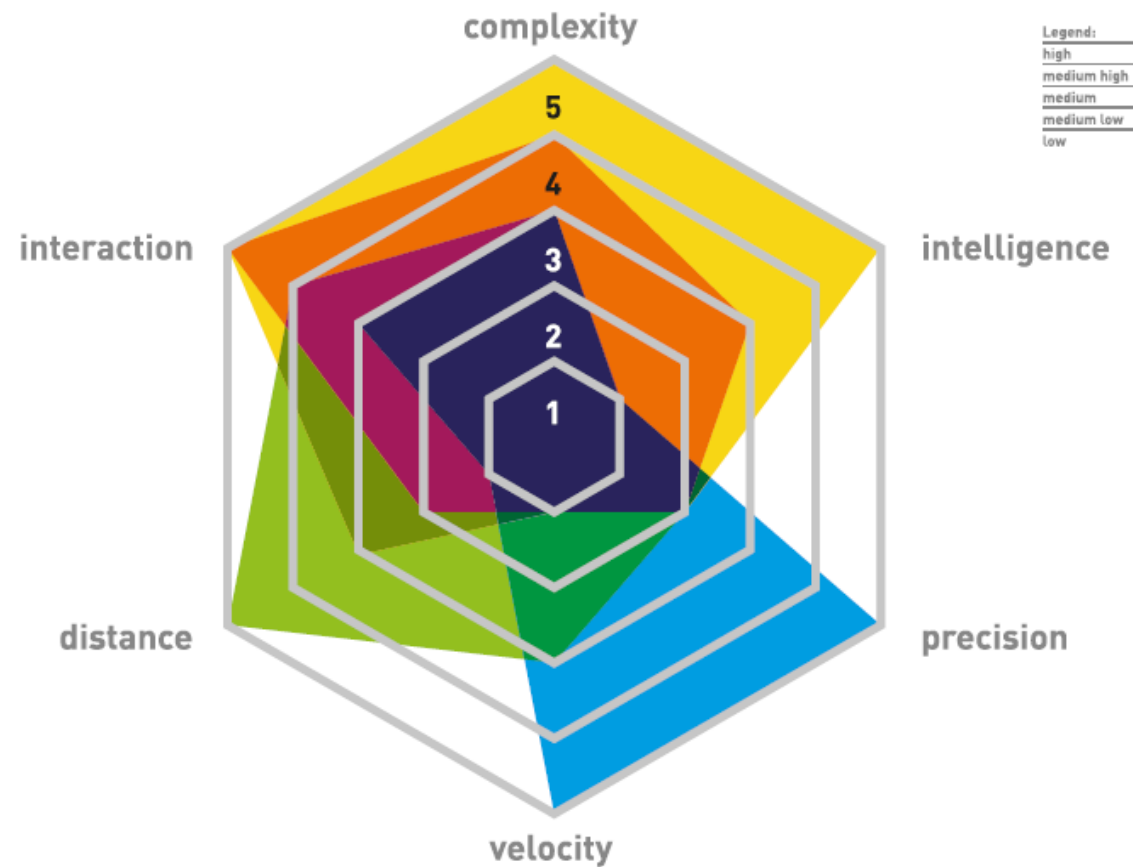


service robot



humanoid robot

MOTION TAXONOMY



1. Taxonomy

Additional Dimension

- Mass: How high is the mass of the moving part (also an energy aspect)?
- ...

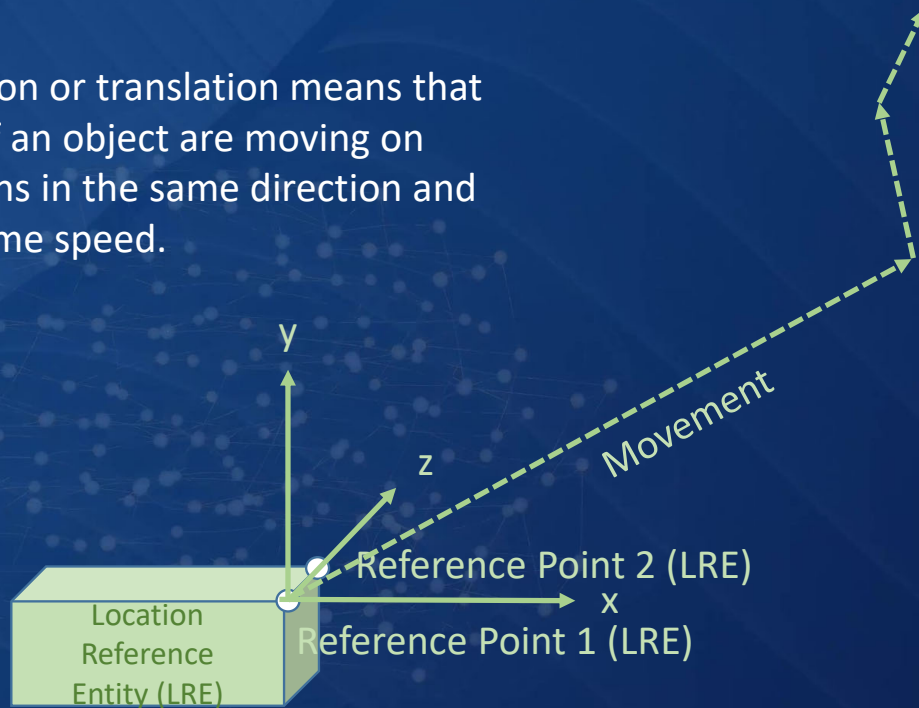


Workshop „Linear and Rotational Motion“

- Where do we only find rotational movement (in which systems)?
- Is linear movement possible without rotational movement? Example?
- Where is linear movement converted into rotational movement?
- Where is rotational movement converted into linear movement?

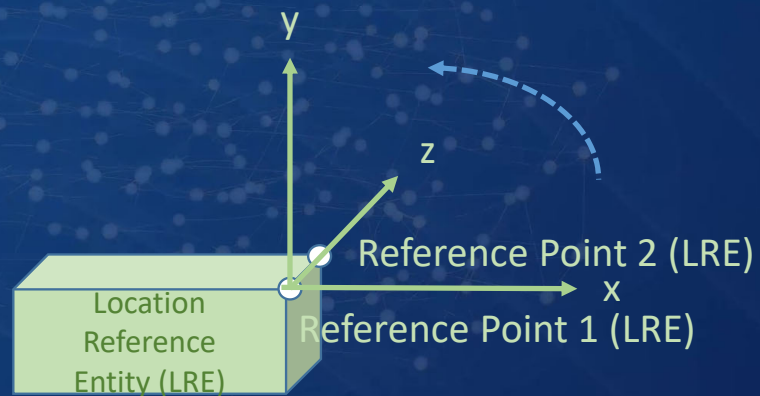
2. Linear Motion

Linear motion or translation means that all points of an object are moving on parallel paths in the same direction and with the same speed.



3. Rotational Motion

Rotational motion means that all points of an object rotate in a circle around one central axis.



4. Part-Whole Logic

Transitivity, Disjointness/Exclusivity

Germany is part of Europe
Bavaria is part of Germany
Würzburg is part of Bavaria
Franz-Horn-Strasse is part of Würzburg
Cairo Building is part of Franz-Horn-Strasse
Staircase is part of Cairo Building
Corridor is part of Cairo Building
K.0.11 is part of Cairo Building
North Wall is part of K.0.11
East Wall is part of K.0.11
South Wall is part of K.0.11
West Wall is part of K.0.11
Door is part of North Wall

...

Transitivity (same time):

if <A> is part of and
 is part of <C>
then <A> is part of <C>

Example:

K.0.11 is part of Europe

Disjointness (same time):

if <A> is part of and
 ! <C> is part of
then ! <A> is part of <C>

Example:

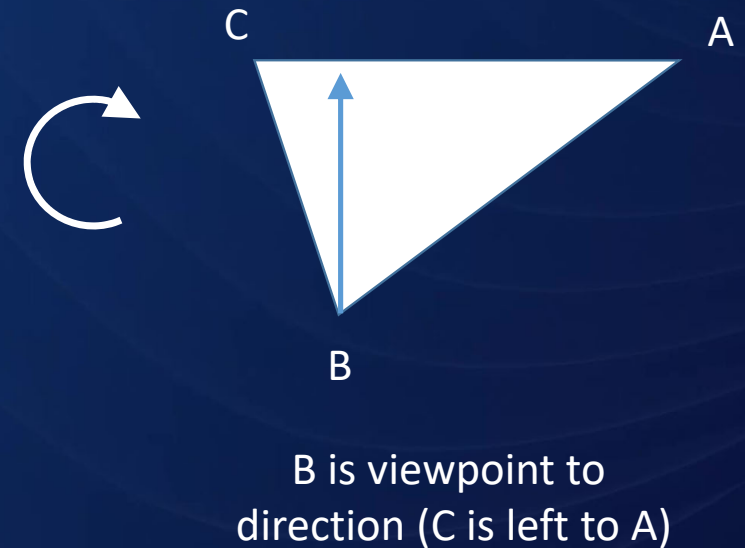
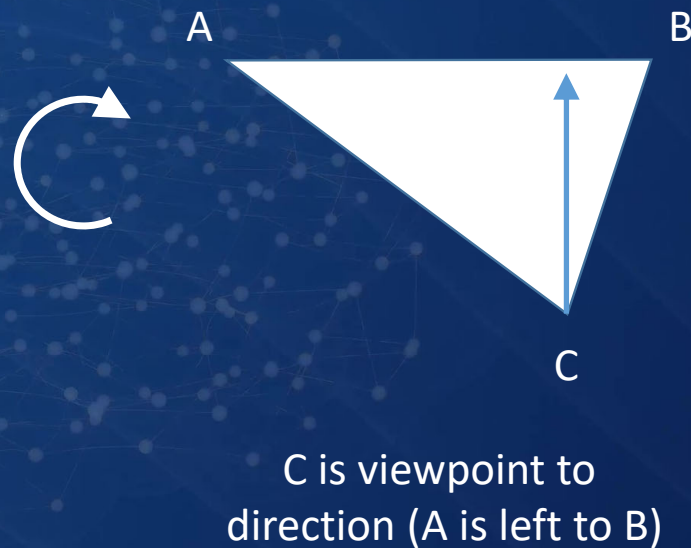
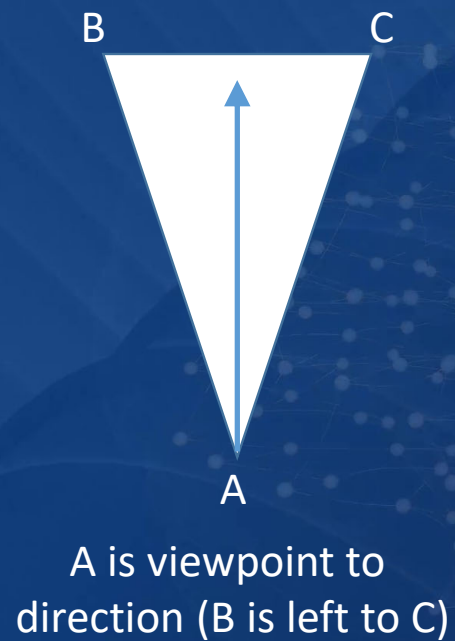
if X: Robert is part of K.0.11 and
 Y: Robert is part of Corridor
then time (X) != time (Y)

Workshop „Rotation“

- What/where is left and what/where is right? Give a definition that is understandable by each human being without using any tools.
- What happens, if you turn yourself for 90 degree and 180 degree? Remains left and right the same?

5. Rotation Logic

Triangle Rotation Paradigm / Left-Right Paradigm



5. Rotation Logic

What is Right?

A Motion Concept: Move our attention (vision, audio) to where the sun rises; when you now look in the direction where the sun is at its highest, you turn right (in the northern hemisphere)

What Helps?

Sun turns right (northern hemisphere)
Clock hands turn right

What is Left?

A Motion Concept: Move our attention (vision, audio) to where the sun sets; when you now look in the direction where the sun is at its highest, you turn left (in the northern hemisphere)

Challenge?

Anchor points

6. Position

Located Entity

C is right to B

X

From my viewpoint X
A is right to B
and
C is right to B

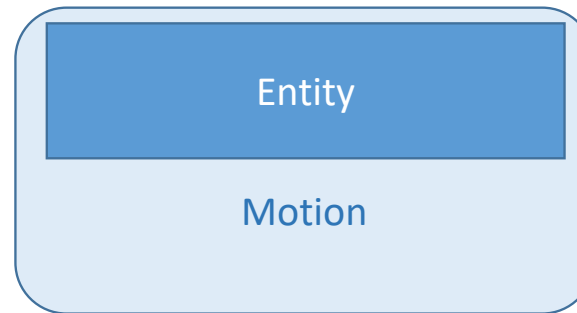
A is right to B

Time and space are mental concepts, that we use to describe what we perceive as motion.

Motion welds time and space together.

A complete description of motion only needs the temporal and the spacial dimensions.

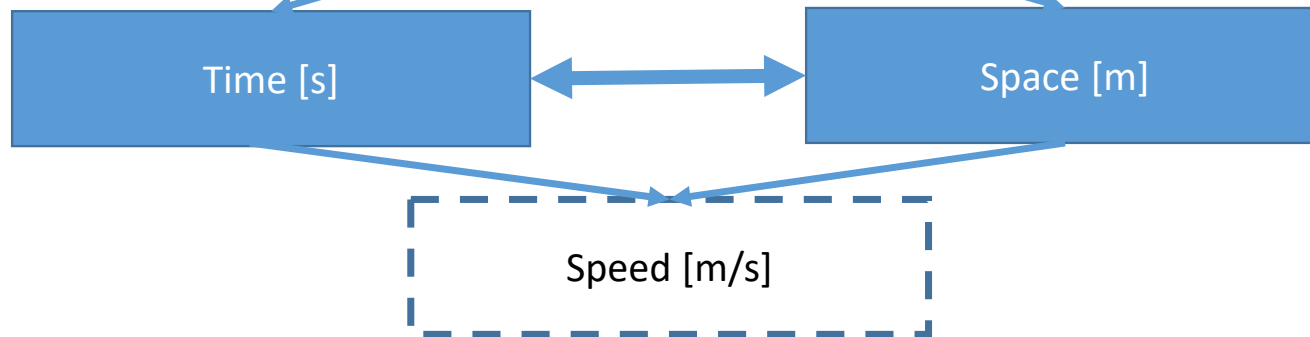
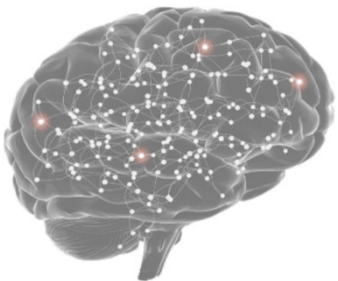
Physical Entities



when?
how long?

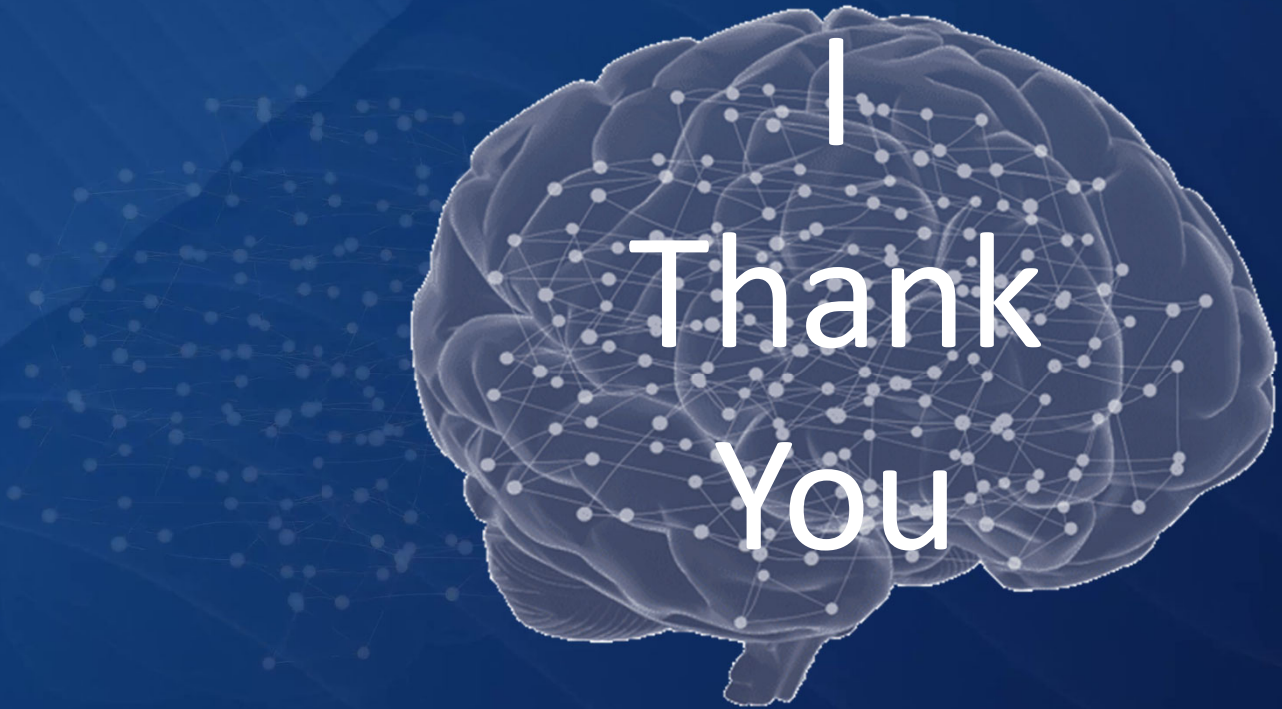
where from?
where to?
how (path, rotation)?

Mental Entities



7. Conclusion

- Motion is the change of location and or orientation of things in the physical world.
- We all are experts in motion when it comes to our body movement, which is based on the rotational motion of our joints.
- Motion in all its facets is very challenging for machines (Physical AI).
- One of the biggest challenges are obstacles that suddenly appear and „destroy“ the action/motion plan which is just executed (see road accidents).
- The left-right paradigm is an easy method to get closer to the way, humans solve the motion challenge, but probably it is not the only one we use.



I
Thank
You